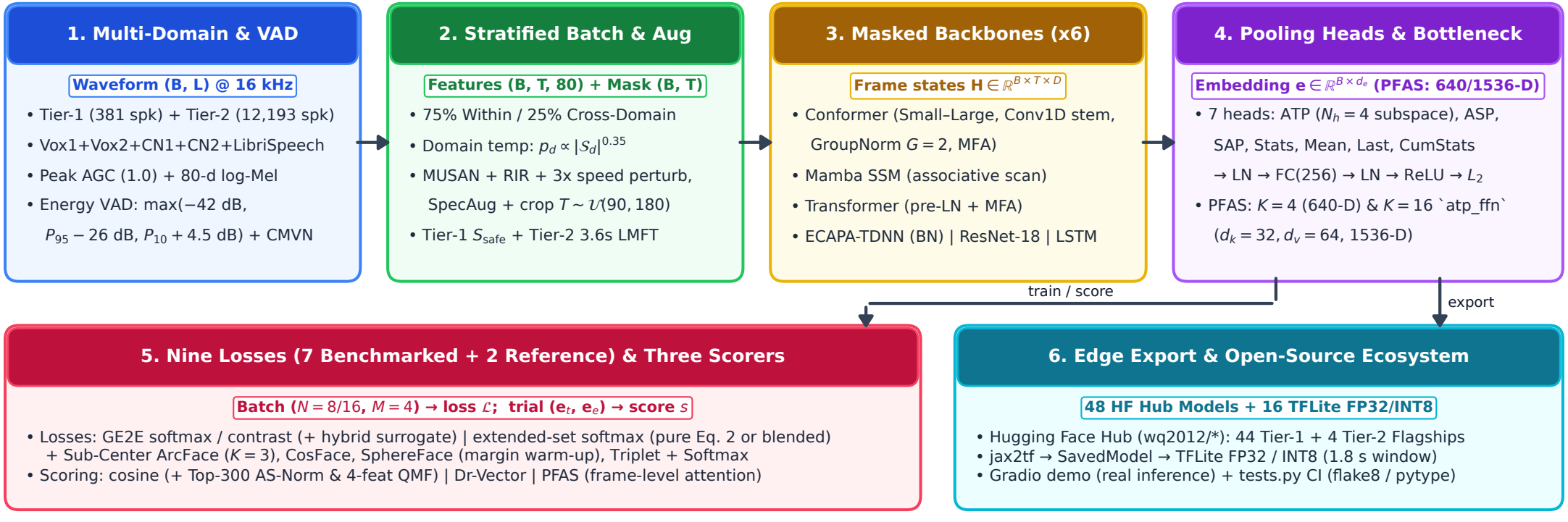

FlaxSpeaker: Multi-Domain Speaker Verification Pipeline (JAX / Flax)
1. Multi-Domain & VAD
Waveform (B, L) @ 16 kHz
• Tier-1 (381 spk) + Tier-2 (12,193 spk)
• Vox1+Vox2+CN1+CN2+LibriSpeech
• Peak AGC (1.0) + 80-d log-Mel
• Energy VAD: max(−42 dB,
$P_{95} - 26$ dB, $P_{10} + 4.5$ dB) + CMVN
2. Stratified Batch & Aug
Features (B, T, 80) + Mask (B, T)
• 75% Within / 25% Cross-Domain
• Domain temp: $p_d \propto |S_d|^{0.35}$
• MUSAN + RIR + 3x speed perturb,
SpecAug + crop $T \sim \mathcal{U}(90, 180)$
• Tier-1 S_{safe} + Tier-2 3.6s LMFT
3. Masked Backbones (x6)
Frame states $\mathbf{H} \in \mathbb{R}^{B \times T \times D}$
• Conformer (Small–Large, Conv1D stem,
GroupNorm $G = 2$, MFA)
• Mamba SSM (associative scan)
• Transformer (pre-LN + MFA)
• ECAPA-TDNN (BN) | ResNet-18 | LSTM
4. Pooling Heads & Bottleneck
Embedding $\mathbf{e} \in \mathbb{R}^{B \times d_e}$ (PFAS: 640/1536-D)
• 7 heads: ATP ($N_h = 4$ subspace), ASP,
SAP, Stats, Mean, Last, CumStats
→ LN → FC(256) → LN → ReLU → L_2
• PFAS: $K = 4$ (640-D) & $K = 16$ `atp_ffn`
($d_k = 32, d_v = 64$, 1536-D)
train / score
export
5. Nine Losses (7 Benchmarked + 2 Reference) & Three Scorers
Batch ($N = 8/16$, $M = 4$) → loss $\mathcal{L}$; trial ($\mathbf{e}_t$, $\mathbf{e}_e$) → score s
• Losses: GE2E softmax / contrast (+ hybrid surrogate) | extended-set softmax (pure Eq. 2 or blended)
+ Sub-Center ArcFace ($K = 3$), CosFace, SphereFace (margin warm-up), Triplet + Softmax
• Scoring: cosine (+ Top-300 AS-Norm & 4-feat QMF) | Dr-Vector | PFAS (frame-level attention)
6. Edge Export & Open-Source Ecosystem
48 HF Hub Models + 16 TFLite FP32/INT8
• Hugging Face Hub (wq2012/*): 44 Tier-1 + 4 Tier-2 Flagships
• jax2tf → SavedModel → TFLite FP32 / INT8 (1.8 s window)
• Gradio demo (real inference) + tests.py CI (flake8 / pytype)